E.7: Antenatal care contact schedules

RECOMMENDATION E.7: Antenatal care models with a minimum of eight contacts are recommended to reduce perinatal mortality and improve women's experience of care.
(Recommended)

Remarks

- The GDG stresses that the four-visit focused ANC (FANC) model does not offer women adequate contact with health-care practitioners and is no longer recommended. With the FANC model, the first ANC visit occurs before 12 weeks of pregnancy, the second around 26 weeks, the third around 32 weeks, and the fourth between 36 and 38 weeks of gestation. Thereafter, women are advised to return to ANC at 41 weeks of gestation or sooner if they experience danger signs. Each ANC visit involves specific goals aimed at improving triage and timely referral of high-risk women and includes educational components *(12)*. However, up-to-date evidence shows that the FANC model, which was developed in the 1990s, is probably associated with more perinatal deaths than models that comprise at least eight ANC visits. Furthermore, evidence suggests that more ANC visits, irrespective of the resource setting, is probably associated with greater maternal satisfaction than less ANC visits.
- The GDG prefers the word "contact" to "visit", as it implies an active connection between a pregnant woman and a health-care provider that is not implicit with the word "visit". In terms of the operationalization of this recommendation, "contact" can be adapted to local contexts through community outreach programmes and lay health worker involvement.
- The decision regarding the number of contacts with a health system was also influenced by the following:
  - evidence supporting improving safety during pregnancy through increased frequency of maternal and fetal assessment to detect problems;
  - evidence supporting improving health system communication and support around pregnancy for women and families;
  - evidence from HIC studies indicating no important differences in maternal and perinatal health outcomes between ANC models that included at least eight contacts and ANC models that included more (11–15) contacts *(203)*;
  - evidence indicating that more contact between pregnant women and knowledgeable, supportive and respectful health-care practitioners is more likely to lead to a positive pregnancy experience.
- Implementation considerations related to this recommendation and the mapping of guideline recommendations to ANC contacts are presented in Chapter 4: Implementation of the ANC guideline and recommendations.

Summary of evidence and considerations

Effects of the FANC model (with four visits) compared with "standard" ANC (with at least eight ANC visits planned) (EB Table E.7)

The evidence on the effects of FANC (the four-visit ANC model) was derived from a Cochrane review on "reduced-visit" ANC models versus "standard" care models (with at least eight ANC visits planned) that included seven RCTs *(203).* Four individual RCTs were conducted in HICs (the United Kingdom and the USA) and three large cluster-RCTs were conducted in LMICs, including one conducted in Argentina, Cuba, Saudi Arabia and Thailand *(204)*, and two conducted in Zimbabwe. The LMIC trials evaluated the FANC model compared with "standard" ANC models that planned for at least eight visits *(12)*. Three cluster-RCTs involving more than 50 000 women contributed data. The median number of visits achieved in the FANC arms of these trials ranged from four to five visits and the median number of visits achieved in the standard ANC arms ranged from four to eight visits.

Maternal outcomes

High-certainty evidence shows that FANC had little or no effect on caesarean section rates (1 trial, 24 526 women; RR: 1.00, 95% CI: 0.89–1.11), and low-certainty evidence suggests that it may make little or no difference to maternal mortality (3 trials, 51 504 women; RR: 1.13, 95% CI: 0.5–2.57).

With regard to maternal satisfaction, outcomes were reported narratively in the review, as data were sparse. In a survey conducted among a subset of women participating in the WHO trial, fewer women were satisfied with the frequency of visits in the FANC model than in the standard model (77.4% versus 87.2%) and women in the FANC model were less likely to be satisfied with the spacing between visits compared with the standard model (72.7% versus 81%). This evidence was not formally graded due to insufficient data.

Fetal and neonatal outcomes

Moderate-certainty evidence indicates that FANC probably increases perinatal mortality compared with "standard" ANC with more visits (3 trials, 51 323 women; RR: 1.15, 95% CI: 1.01–1.32). Based on this RR, the illustrative impact on perinatal mortality rates are shown in Box 4.

Moderate-certainty evidence indicates that FANC probably has little or no effect on preterm birth (3 trials, 47 094 women; RR: 0.99, 95% CI: 0.91–1.08) and low birth weight (3 trials, 46 220 women; RR: 1.04, 95% CI: 0.97–1.12) compared with "standard" ANC. In addition, low-certainty evidence suggests that FANC probably makes little or no difference to SGA (3 trials, 43 094 women; RR: 1.01, 95% CI: 0.88–1.17).

Additional considerations

- The GDG noted that the review authors explored reasons for the effect on perinatal mortality and the effect persisted in various exploratory analyses.
- In 2012, the WHO undertook a secondary analysis of perinatal mortality data from the WHO FANC trial *(205)*. This secondary analysis, which included 18 365 low-risk and 6160 high-risk women, found an increase in the overall risk of perinatal mortality between 32 and 36 weeks of gestation with FANC compared with "standard" ANC in both low- and high-risk populations.
- It is not clear whether the philosophy of the FANC approach, with regard to improving quality of care at each ANC visit, was implemented effectively in the trials. However, if this element is neglected, a poorly executed FANC model may then simply represent reduced health provider contact, and a reduced opportunity to detect risk factors and complications, and to address women's concerns.
- The GDG panel considered unpublished findings of a two-year audit of perinatal mortality from the Mpumalanga region of South Africa that has implemented the FANC model *(206)*. The audit from September 2013 to August 2015 comprised data of 149 308 births of neonates weighing more than 1000 g, among which there were 3893 perinatal deaths (giving a PMR of 24.8 per 1000 births). Stillbirth risk was plotted according to gestational age and three peaks in the occurrence of stillbirths were noted, one at around 31 weeks of gestation, another at around 37 weeks, and the third occurring at 40 weeks or more. When these data were compared with stillbirth data from another South Africa province, which uses a model of ANC that includes fortnightly ANC visits from 28 weeks of gestation, the latter showed a gradual rise in the overall stillbirth risk from 28 weeks, with a single (and lower) peak at 40 weeks or more, i.e. no additional peaks at 30 and 37 weeks. These data are consistent with those from the secondary analysis of the WHO trial and suggest that additional visits in the third trimester may prevent stillbirths.
- The GDG also considered the evidence from the Cochrane review on reduced visit ANC models of at least eight visits versus "standard" ANC models with 11–15 visits from four RCTs in HICs *(203)*. Low-certainty evidence suggested that the reduced-visit model (with at least eight visits) may be associated with increased preterm birth

Box 4: Illustration of the impact of focused ANC (FANC) on perinatal mortality rates (PMR)

Assumed PMR ("Standard" ANC)	Illustrative PMR[a] (FANC model)	Absolute increase in perinatal deaths
10 deaths per 1000 births	12 deaths per 1000 births (10–13 deaths)	2 deaths per 1000 births (0–3 deaths)
25 deaths per 1000 births	29 deaths per 1000 births (25–33 deaths)	4 deaths per 1000 births (0–8 deaths)
50 deaths per 1000 births	58 deaths per 1000 births (50–66 deaths)	8 deaths per 1000 births (0–16 deaths)

a Based on RR: 1.15, 95% CI: 1.01–1.32.

(3 trials; RR: 1.24, 1.01–1.52), but no other important effects on health outcomes were noted. In general, however, evidence from these individual studies also suggests that the reduced-visit models may be associated with lower women's satisfaction.

- The GDG considered unpublished evidence from four country case studies (Argentina, Kenya, Thailand and the United Republic of Tanzania) where the FANC model has been implemented *(207)*. Provider compliance was noted to be problematic in some settings, as were shortages of equipment, supplies and staff. Integration of services was found to be particularly challenging, especially in settings with a high prevalence of endemic infections (e.g. malaria, TB, sexually transmitted infections, helminthiasis). Guidance on implementation of the FANC model in such settings was found to be inadequate, as was the amount of time allowed within the four-visit model to provide integrated care.
- Findings on provider compliance from these case studies are consistent with published findings from rural Burkina Faso, Uganda and the United Republic of Tanzania *(208)*. Health-care providers in this study were found to variably omit certain practices from the FANC model, including blood pressure measurement and provision of information on danger signs, and to spend less than 15 minutes per ANC visit. Such reports suggest that fitting all the components of the FANC model into four visits is difficult to achieve in some low-resource settings where services are already overstretched. In addition, in low-resource settings, when the target is set at four ANC visits, due to the various barriers to ANC use, far fewer than four visits may actually be achieved.
- Programmatic evidence from Ghana and Kenya indicates similar levels of satisfaction between FANC and standard ANC, with sources of dissatisfaction with both models being long waiting times and costs associated with care *(209, 210)*.
- Emotional and psychosocial needs are variable and the needs of vulnerable groups (including adolescent girls, displaced and war-affected women, women with disabilities, women with mental health concerns, women living with HIV, sex workers, ethnic and racial minorities, among others) can be greater than for other women. Therefore, the number and content of visits should be adaptable to local context and to the individual woman.

Values

See "Women's values" at the beginning of section 3.E: Background (p. 86).

Resources

Two trials evaluated cost implications of two models of ANC with reduced visits, one in the United Kingdom and one in two LMICs (Cuba and Thailand). Costs per pregnancy to both women and providers were lower with the reduced visits models in both settings. Time spent accessing care was also significantly shorter with reduced visits models. In the United Kingdom trial, there was an increase in costs related to neonatal intensive care unit stays in the reduced visit model.

Equity

Preventable maternal and perinatal mortality is highest among disadvantaged populations, which are at greater risk of various health problems, such as nutritional deficiencies and infections, that predispose women to poor pregnancy outcomes. This suggests that, in LMICs, more and better quality contact between pregnant women with health-care providers would help to address health inequalities.

Acceptability

Evidence from high-, medium- and low-resource settings suggests that women do not like reduced visit schedules and would prefer more contact with antenatal services (moderate confidence in the evidence) *(22)*. Women value the opportunity to build supportive relationships during their pregnancy (high confidence in the evidence) and for some women, especially in LMIC settings, the reduced visit schedule may limit their ability to develop these relationships, both with health-care professionals and with other pregnant women (low confidence in the evidence). In some low-income settings where women rely on husbands or partners to financially support their antenatal visits, the reduced visit schedule limits their ability to procure additional finance (low confidence in the evidence). However, the reduced visit schedule may be appreciated by some women in a range of LMIC settings because of the potential for cost savings, e.g. loss of domestic income from extra clinic attendance and/or associated travel costs (low confidence in the evidence). Indirect evidence also suggests that women are much more likely to engage with antenatal services if care is provided by knowledgeable, kind health-care professionals who have the time and resources to deliver genuine woman-centred care, regardless of the number of visits (high confidence in the evidence). Specific evidence from providers relating to reduced visit schedules or the adoption of FANC is sparse and, in some LMICs, highlights concerns around the availability of equipment and resources, staff shortages and inadequate training – issues that are pertinent to all models of ANC delivery in low-resource settings.

Feasibility

Qualitative evidence suggests that some providers in LMICs feel that the reduced visit schedule is a more efficient use of staff time and is less likely to deplete limited supplies of equipment and medicine (moderate confidence in the evidence) *(45)*. Programme reports from Ghana and Kenya stress that inadequate equipment, supplies, infrastructure and training may hamper implementation *(209, 210)*. Providers have also raised concerns about the difficulty of incorporating all of the FANC components into relatively short appointments, especially in LMICs (Burkina Faso, Uganda and the United Republic of Tanzania) where services are already stretched *(208, 211)*.

4. Implementation of the ANC guideline and recommendations: introducing the 2016 WHO ANC model

The ultimate goal of this guideline and its recommendations is to improve the quality of ANC and to improve maternal, fetal and newborn outcomes related to ANC. These ANC recommendations need to be deliverable within an appropriate model of care that can be adapted to different countries, local contexts and the individual woman. With the contributions of the members of the Guideline Development Group (GDG), WHO reviewed existing models of delivering ANC with full consideration of the range of interventions recommended within this guideline (Chapter 3). Recommendation E.7 states that "Antenatal care models with a minimum of eight contacts are recommended to reduce perinatal mortality and improve women's experience of care"; taking this as a foundation, the GDG reviewed how ANC should be delivered in terms of both the timing and content of each of the ANC contacts, and arrived at a new model – the 2016 WHO ANC model – which replaces the previous four-visit focused ANC (FANC) model. For the purpose of developing this new ANC model, the ANC guideline recommendations were mapped to the eight contacts based on the evidence supporting each recommendation and the optimal timing of delivery of the recommended interventions to achieve maximal impact.

The 2016 WHO ANC model recommends a minimum of eight ANC contacts, with the first contact scheduled to take place in the first trimester (up to 12 weeks of gestation), two contacts scheduled in the second trimester (at 20 and 26 weeks of gestation) and five contacts scheduled in the third trimester (at 30, 34, 36, 38 and 40 weeks). Within this model, the word "contact" has been used instead of "visit", as it implies an active connection between a pregnant woman and a health-care provider that is not implicit with the word "visit". It should be noted that the list of interventions to be delivered at each contact and details about where they are delivered and by whom (see Table 2) are not meant to be prescriptive but, rather, adaptable to the individual woman and the local context, to allow flexibility in the delivery of the recommended interventions. Different to the FANC model, an additional contact is now recommended at 20 weeks of gestation, and an additional three contacts are recommended in the third trimester (defined as the period from 28 weeks of gestation up to delivery), since this represents the period of greatest antenatal risk for mother and baby (see Box 5). At these third-trimester contacts, ANC providers should aim to reduce preventable morbidity and mortality through systematic monitoring of maternal and fetal well-being, particularly in relation to hypertensive disorders and other complications that may be asymptomatic but detectable during this critical period.

Box 5: Comparing ANC schedules

WHO FANC model	2016 WHO ANC model
First trimester	
Visit 1: 8–12 weeks	Contact 1: up to 12 weeks
Second trimester	
Visit 2: 24–26 weeks	Contact 2: 20 weeks Contact 3: 26 weeks
Third trimester	
Visit 3: 32 weeks Visit 4: 36–38 weeks	Contact 4: 30 weeks Contact 5: 34 weeks Contact 6: 36 weeks Contact 7: 38 weeks Contact 8: 40 weeks
Return for delivery at 41 weeks if not given birth.	

If the quality of ANC is poor and women's experience of it is negative, the evidence shows that women will not attend ANC, irrespective of the number of recommended contacts in the ANC model. Thus, the overarching aim of the 2016 WHO ANC model is to provide pregnant women with respectful, individualized, person-centred care at every contact, with implementation of effective clinical practices (interventions and tests), and provision of relevant and timely information, and psychosocial and emotional support, by practitioners with good clinical and interpersonal skills within a well functioning health system. Effective implementation of ANC requires a health systems approach and strengthening focusing on continuity of care, integrated service delivery, availability of supplies and commodities and empowered health-care providers.

There are many different ways for health system planners to optimize ANC delivery by employing a range of strategies that can improve the utilization and quality of ANC. The health system recommendations in this guideline have focused mainly on those strategies that address continuity of care, and improve communication with, and support for, women (Recommendations E.1–E.4). The recommendations on task shifting and recruitment of staff (Recommendations E.5.1, E.5.2 and E.6) are also important, as provider experience and attitudes have an impact on the capacity of health systems to deliver quality ANC; barriers to provider recruitment and job satisfaction will need to be addressed to successfully implement this guideline. Such barriers have been shown to be significant in LMICs, and can prevent the provision of quality midwifery care *(212)*. In addition to improving the quality of care, these health system recommendations are intended to encourage health system planners to operationalize the recommended eight ANC contacts in ways that are feasible in the local context.

Table 2 shows the WHO ANC guideline recommendations mapped to the eight recommended contacts, thus presenting a summary framework for the 2016 WHO ANC model in support of a positive pregnancy experience. This table does not include good clinical practices, such as measuring blood pressure, proteinuria and weight, checking for fetal heart sounds, which would be included as part of an implementation manual aimed at practitioners. Practices that are not recommended have been included in the table for informational purposes and highlighted in grey. Context-specific recommendations for which rigorous research is required before they can be considered for implementation have not been mapped to the schedule of contacts.

Any intervention that is missed at an ANC contact, for any reason, should in principle be included at the next contact. Effective communication should be facilitated at all ANC contacts, to cover: presence of any symptoms; promotion of healthy pregnancies and newborns through lifestyle choices; individualized advice and support; timely information on tests, supplements and treatments; birth-preparedness and complication-readiness planning; postnatal family planning options; and the timing and purpose of ANC contacts. Topics for individualized advice and support can include healthy eating, physical activity, nutrition, tobacco, substance use, caffeine intake, physiological symptoms, malaria and HIV prevention, and blood test results and retests. Communication should occur in a respectful, individualized and person-centred way. An effective referral system and emergency transport are also essential components of this ANC model.

Within the 2016 WHO ANC model, there are two opportunities to arrange a single early ultrasound scan (i.e. before 24 weeks of gestation): either at the first contact (up to 12 weeks of gestation) or at the second contact (20 weeks). The GDG suggests this pragmatic approach in order to increase the proportion of pregnancies with accurate gestational age assessments, especially in settings where ANC utilization is historically low; lack of accurate gestational age assessment can compromise the diagnosis and/or management of complications (such as preterm birth and pre-eclampsia). It is important to highlight that the frequency and exact timing of some of these ANC practices and interventions – especially related to malaria, tuberculosis and HIV – may need to be adapted, based on the local context, population and health system. Please refer to Box 6 at the end of this chapter for considerations related to the adoption, scale-up and implementation of the 2016 WHO ANC model.

The GDG agreed that implementation of the 2016 WHO ANC model should not wait for a large multicentre trial to be conducted to determine the optimal number of contacts, or the impact of the additional recommended interventions, such as ultrasound, on pregnancy outcomes, resources, equity and the other domains; rather, following implementation of the model, it should be subject to ongoing monitoring and evaluation. It should be remembered that the four-visit model has significantly increased stillbirth risk compared to standard models with eight or more contacts. Understandably, policy-makers and health-care providers might feel that an increase in the number of ANC contacts with an emphasis on quality of care will increase the burden on already overstretched health systems. However, the GDG agreed that there is likely to be little impact on lives saved or improved without substantial investment in improving the quality of ANC services provided in LMICs. International human rights law requires that States use "maximum available resources" to realize economic, social and cultural rights, which includes women's rights to sexual and reproductive health *(1).* Ensuring that women's rights to sexual and reproductive health are supported requires meeting standards with regard to the availability, accessibility, acceptability and quality of health-care facilities, supplies and services *(1)*. Specifically, in addition to other health system strengthening initiatives, investment is urgently needed to address the shortage and training of midwives and other health-care providers able to offer ANC. Such investment should be considered a top priority as quality health care around pregnancy and childbirth has far-reaching benefits for individuals, families, communities and countries.